Intelligent Manufacturing Engineering

	Normative 4-year Degree
Major	Intelligent Manufacturing Engineering 智能製造工程學
Degree / Award Title	Bachelor of Engineering in Intelligent Manufacturing Engineering 工學士 (智能製造工程學)
Offering Academic Unit	Department of Systems Engineering
Normal Period of Study	4 years
Maximum Period of Study	8 years
Credit Units Required for Graduation	121 credit units

Note: The following curriculum information is subject to periodic review and changes.

GE Requirements (31 credit units) Catalogue Term : Semester A 2025/26

		Credit Units
GE University Requirements	GE1401 University English	3
	GE2410 English for Engineering	3
	GE1501 Chinese Civilisation - History and Philosophy	3
	GE1601 Whole-Person Development	1
GE Distributional Requirements	Take at least 1 course from each of the three distributional areas: Area 1: Arts and Humanities Area 2: Study of Societies, Social and Business Organisations Area 3: Science and Technology Visit the GE website for more course information.	12

College-specified Courses	MA1200 Calculus and Basic Linear Algebra I *Or* MA1300 Enhanced Calculus and Linear Algebra I	3
	MA1201 Calculus and Basic Linear Algebra II *Or* MA1301 Enhanced Calculus and Linear Algebra II	3
	SYE2066 Professional Engineering Practice	3

College / School Requirements (6 credit units) Catalogue Term : Semester A 2025/26

Course Code	Course Title	Credit Units
CS1302	Introduction to Computer Programming	3
PHY1201	General Physics I	3

Major Requirements (81 credit units) Catalogue Term : Semester A 2025/26

1. Core Courses (66 credit units)

Course Code	Course Title	Credit Units
SYE2010	Fundamental Engineering Analysis and Design for Manufacturing Engineers I	3
SYE2011	Fundamental Engineering Analysis and Design for Manufacturing Engineers II	3
SYE2046	Numerical Computation for Manufacturing and Systems Engineers	3
SYE2339	Smart City - a Systems Engineering Perspective	3
MSE2102	Introduction to Materials Science and Engineering	3
SYE2016	Manufacturing Engineering Workshop	0
SYE2100	Engineering Statistics and Experimentation	3
MNE3007	CAD/CAM	3
MNE3046	Automation Technology	3
MNE3119	Manufacturing Technology	3
SYE3003	Design and Analysis of Manufacturing Processes and Systems	3
SYE3004	Production Planning and Control	3
SYE3060	Operations Research	3
SYE3102	Quality Engineering	3
SYE4001	Digital Manufacturing and Operations	3
SYE4002	Fundamentals of Advanced Packaging	3
SYE4003	Artificial Intelligence and Advanced Technology in Manufacturing and Operations	3
SYE4005	Industrial Data and Manufacturing Analytics	3
SYE4006	Semiconductor Manufacturing and Process Control	3
SYE4036	Systems Modelling and Optimization	3

SYE4064	Reliability Engineering	3

Select **ONE** from the following:

Course Code	Course Title	Credit Units	Remarks
SYE4068	Final Year Project	6	A course to fulfil the internship/consultancy project/research project requirement for normative 4-year degree students
SYE4068C	Final Year Project	6	A course to fulfil the internship/consultancy project/research project requirement for normative 4-year degree students
SYE4116	Capstone Project II	6	A course to fulfil the internship/consultancy project/research project requirement for normative 4-year degree students

2. Electives (15 credit units)

Course Code	Course Title	Credit Units
EE1002	Principles of Electrical Engineering	3
EE2000	Logic Circuit Design	3
EE2005	Electronic Devices and Circuits	3
MNE2109	Engineering Mechanics	3
SDSC2004	Data Visualization	3
EE3009	Data Communications and Networking	3
EE3315	Internet Technology	3
PHY3202	Modern Physics	3
SYE3018	Control Systems	3
SYE3026	Contemporary Human Factors for Industry 4.0	3
SYE3101	Basic Methodologies and Tools for Risk Engineering	3
SYE3116	Capstone Project I	3
MNE4032	Robotics and Machine Vision	3
MNE4125	Principles and Transport Process Fundamentals of Semiconductor Manufacturing	3
MSE4178	Nanostructures and Nanotechnology	3
SYE4007	eLogistics and Supply Chain Management	3
SYE4008	Computer-Aided Problem Solving for Manufacturing and Systems Engineers	3
SYE4024	Project Management	3
SYE4027	Occupational Safety for Intelligent Manufacturing Systems	3

SYE4035	Quality and Environmental System and Management	3
SYE4047	Directed Studies	3
SYE4059	Non-Destructive Testing Technologies for Process Monitoring and Inspection	3
SYE4103	Decision Analysis and Risk Management	3
SYE4108	Product Development and Innovation	3

For Undergraduate plus Master's Degree Programmes only

A maximum of 9 credit units of P5 and/or P6 courses can be used to fulfill the MSc degree requirements.

Course Code	Course Title	Credit Units
SYE5009	Industrial Marketing Management for Engineers	3
SYE5010	Engineering Management Principles and Concepts	3
SYE6009	Project Management	3
SYE6012	Technological Innovation and Entrepreneurship	3
SYE6015	Supply Chain Management	3
SYE6053	Business Process Improvement and Innovation	3
SYE6102	Managerial Decision-making Systems with Artificial Intelligence	3
SYE6103	Financial Engineering for Engineering Managers	3
SYE6106	Intelligent Manufacturing for Engineering Managers	3
SYE6107	Contemporary Occupational Safety and Health Management	3
SYE6110	Data Analysis and Artificial Intelligence for Systems Engineering	3
SYE6111	Transportation and Logistics Management	3

3. Optional Courses

Course Code	Course Title	Credit Units	Remarks
FS2002	Industrial Attachment Scheme	3	Minimum 6 weeks
FS3002	Industrial Attachment Scheme	3	Minimum 6 weeks
FS4001	Co-operative Education Scheme (CES)	8	Internship (8 to 12 months)
FS4002	Industrial Attachment Scheme	3	Minimum 6 weeks